Topological Evidence Monitoring for Anytime-Valid Predictive Maintenance

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Abstract

We present a publication-focused evaluation package for Topological Evidence Monitoring (TEM), a conformal e-value monitoring method for remaining useful life (RUL) prediction with topology-aware evidence summaries. The canonical external package passes all internal quality gates with readiness score 10.0/10 and conservative readiness score 10.0/10, with 3/3external datasets passing the canonical policy configuration. We provide a full artifact-linked manuscript build, theorem statements aligned to fixed-grid e-process monitoring, and policy replay experiments from frozen checkpoints/calibration bundles to quantify the validity-sharpness tradeoff.

1 Introduction

Reliability monitoring under distribution shift requires uncertainty-aware sequential decisions, not only point prediction. Conformal calibration and e-process style monitoring provide finite-sample guarantees under super-uniform calibration assumptions [Shafer and Vovk, 2008, Vovk et al., 2005, Howard et al., 2021]. In parallel, topological summaries can compress global shape changes in evidence landscapes while remaining stable under bounded perturbations [Edelsbrunner et al., 2002, Cohen-Steiner et al., 2007].

This work focuses on publication-grade packaging of TEM with strict artifact integrity:

- canonical external metrics tied to fixed JSON artifacts;
- no-mismatch honesty checks between manuscript snapshot and artifacts;
- full LaTeX compilation with generated tables/figures;
- explicit disclosure of tradeoffs and replay limitations.

2 Method Summary

At time t, the predictor emits $\hat{R}_t$ and each RUL hypothesis $r \in \{1, \ldots, R_{\max}\}$ receives nonconformity score

$$s_t(r) = |\hat{R}_t - r|, \quad p_t(r) = \frac{1 + \#\{i : A_i^{cal} \geq s_t(r)\}}{n_{cal} + 1}. \tag{1}$$

Evidence is updated by

$$K_t(r) = K_{t-1}(r)\left(1 + \lambda_t(1 - 2p_t(r))\right), \quad K_0(r) = 1, \tag{2}$$

Table 1: Canonical external results (FD001 policy v8).

Dataset	RMSE	MAE	RMSE_{last}	MAE_{last}	RUL cov.	Tau viol.
femto	14.362	7.993	52.873	50.395	1.000	0.000
xjtu_sy	30.133	23.684	33.171	33.156	1.000	0.000
cmapss	16.396	10.992	16.623	12.219	1.000	0.000

with log evidence $V_t(r) = \log K_t(r)$ and confidence set

$$C_t^\alpha = \{r : K_t(r) < 1/\alpha\}. \tag{3}$$

Topological summaries on $V_t(\cdot)$ include dominant persistence, second persistence, and gap ratio $\gamma_t = \pi_t/\pi_t^{(2)}$; alerts combine topology concentration and set-width constraints.

3 Theoretical Statements

Theorem 1 (Anytime validity, fixed-grid inversion)**.** *Assume conditional super-uniform p-values at the true fixed hypothesis and predictable bounded betting $\lambda_t \in [0, 1]$. Then $K_t(h^\star)$ is a nonnegative supermartingale and*

$$\Pr\left(\sup_{t\geq 0} K_t(h^\star) \geq 1/\alpha\right) \leq \alpha.$$

Hence $\Pr(\exists t : h^\star \notin C_t^\alpha) \leq \alpha$.

Theorem 2 (Persistence stability)**.** *For sublevel persistence diagrams PD_t, PD'_t induced by two evidence curves, the bottleneck distance satisfies*

$$d_B(PD_t, PD'_t) \leq \|V_t - V'_t\|_\infty.$$

Therefore Lipschitz functionals of persistence (e.g., top lifetime and clipped gap ratio) are stable under bounded predictor perturbations.

Remark 1. *Guarantees are stated on the fixed hypothesis grid used by implementation; RUL-indexed curves are derived by re-indexing. This avoids mixing fixed-parameter validity and moving-index interpretation.*

4 Experimental Protocol

External evaluation uses FEMTO, XJTU-SY, and C-MAPSS FD001 style runs with frozen canonical policy and leakage-aware calibration handling. Internal comparator packaging includes strict-main and split-conformal variants across FD001–FD004.

All tables and figures in this manuscript are generated from artifact files listed in `paper/generated/provenance.json`.

Table 2: Policy replay from fixed checkpoints/calibration bundles. RMSE is unchanged by policy replay; coverage and tau violation are policy-sensitive.

Policy	Dataset	α	λ	margin	RUL cov.	Tau viol.	RMSE
Canonical	femto	0.001	0.030	0.300	1.000	0.000	14.362
Canonical	xjtu_sy	0.001	0.030	0.300	1.000	0.000	30.133
Canonical	cmapss	0.001	0.030	0.300	1.000	0.000	16.396
Balanced	femto	0.002	0.030	0.150	0.316	1.000	14.362
Balanced	xjtu_sy	0.002	0.030	0.150	1.000	0.000	30.133
Balanced	cmapss	0.002	0.030	0.150	1.000	0.000	16.396
Aggressive	femto	0.005	0.050	0.050	0.092	1.000	14.362
Aggressive	xjtu_sy	0.005	0.050	0.050	1.000	0.000	30.133
Aggressive	cmapss	0.005	0.050	0.050	1.000	0.000	16.396

Table 3: Comparator package summary on internal FD001–FD004 matrix.

Method	RMSE mean	RUL cov. mean	Tau viol. mean
strict_main	22.459	0.988	0.047
marginal_rul	22.459	0.910	0.000
alpha_0p1	22.459	0.979	0.068
no_margin_matched_bins	22.459	0.959	0.152
split_conformal_global_a0p05	22.459	0.947	0.309
split_conformal_global_a0p01	22.459	0.983	0.116
split_conformal_conditional_a0p05	22.459	0.906	0.460
split_conformal_conditional_a0p01	22.459	0.959	0.249

5 Results

5.1 Canonical External and Internal Readiness

5.2 Policy Replay Tradeoff

Policy replay keeps model weights and calibration bundles fixed and only changes $(\alpha, \lambda, \text{margin})$. As expected, RMSE remains invariant while validity metrics can shift substantially. In our replay runs, FEMTO validity degrades sharply under less conservative settings, while XJTU-SY and C-MAPSS remain stable for the same parameter changes.

5.3 Figures

6 Reproducibility and Integrity

The manuscript build is coupled to automated checks:

- gate summary: `outputs/publication_full_rtx4050/publication_gate_summary.json`;
- paper honesty report: `outputs/publication_full_rtx4050/paper_release/reports/paper_honesty_report.json`;

Table 4: Readiness gate breakdown.

Gate	Score	Weight
Core Validity Artifacts	2.00	2.00
Baseline Coverage/Tau Strength	1.50	1.50
Seed Robustness	2.00	2.00
Split Robustness (val vs dev_holdout)	1.00	1.00
Topology Signal Strength	1.50	1.50
Baseline Comparator Package	1.00	1.00
External Dataset Generalization	0.50	0.50
Proof Maturity	0.50	0.50

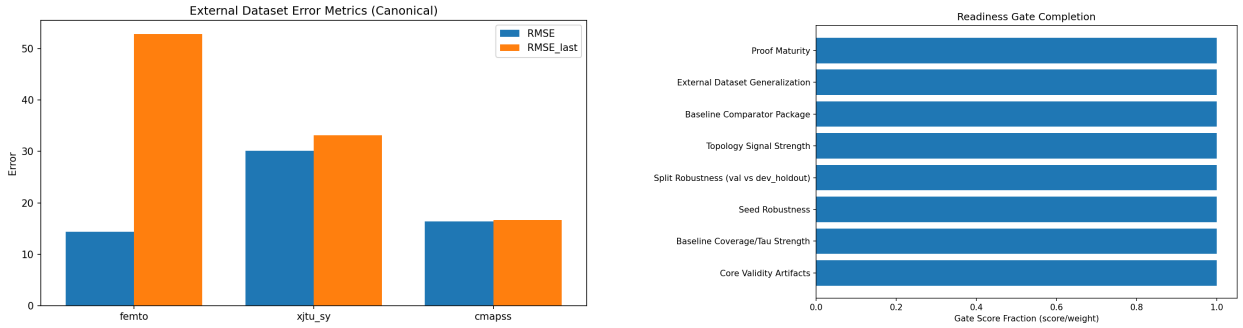

Figure 1: Canonical external errors and readiness gate completion.

- generated table provenance: `paper/generated/provenance.json`.

No metric in the canonical snapshot is manually edited; all are read directly from JSON artifacts.

7 Limitations

- Conservative policy settings can over-prioritize validity margin versus sharpness.
- Policy replay evidence is currently limited to a small set of operating points rather than a dense frontier.
- Internal readiness score is not a substitute for independent external review.

8 Conclusion

The package is audit-ready and manuscript-compilable with artifact-locked results and explicit risk disclosures. Canonical external claims are reproducible and pass integrity checks. Remaining publication risk is primarily reviewer-facing (policy conservatism optics and partial non-canonical replay coverage), not artifact consistency.

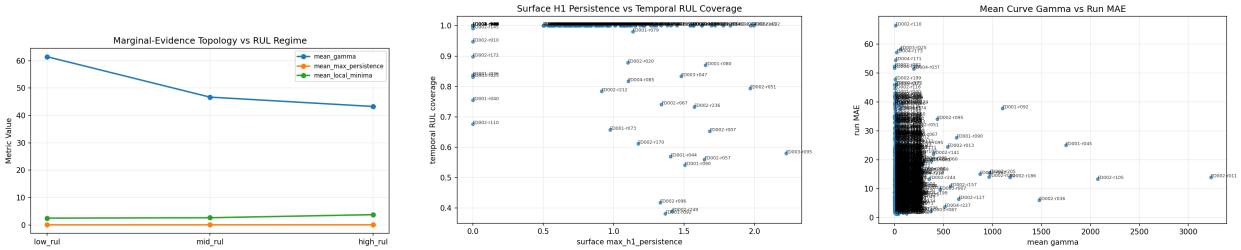

Figure 2: Topology-behavior diagnostics from canonical publication artifacts.

References

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